ARAVIND NAIDU YALLA

AI ENGINEER • GENERATIVE AI • LLMs • RAG

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PROFESSIONAL SUMMARY

AI Engineer at People Media Factory with 1 year of professional experience developing production-grade AI-powered applications. Specialized in Python, FastAPI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector embeddings, semantic search, and agentic architectures. Proven track record in orchestrating multi-document ingestion pipelines, vector databases (Qdrant), and low-latency inference APIs.

TECHNICAL SKILLS

AI & Core LLM:	Generative AI, Large Language Models (LLMs), RAG, Agentic AI, Embeddings, Semantic Search, Ollama, Hugging Face
Backend & APIs:	Python, FastAPI, RESTful APIs, Asynchronous Programming, Microservices Architecture
Databases & Vector:	Qdrant (Vector Database), PostgreSQL, Similarity Search, Indexing & Chunking
Tools & Version Control:	Git, GitHub, Docker, Linux, OCR Text Extraction Pipelines, Prompt Engineering

PROFESSIONAL EXPERIENCE

People Media Factory — AI Engineer — Hyderabad, India | Oct 2025 – Present

Transitioned from AI Intern to Full-time AI Engineer

- Architected and deployed production-grade AI-powered applications, integrating state-of-the-art LLMs with scalable Python FastAPI backends.
- Engineered end-to-end Retrieval-Augmented Generation (RAG) systems incorporating chunking, OCR pipelines, and vector database indexing.
- Spearheaded AI experimentation, prompt tuning, and model evaluation to maximize generation relevance and factual grounding.
- Built resilient REST APIs and microservice endpoints supporting enterprise media and production workflows.

KEY PROJECTS

Enterprise RAG System | *Document QA Intelligence* — Python, FastAPI, Ollama, Qdrant, Hugging Face, PostgreSQL

- Built an enterprise Retrieval-Augmented Generation system enabling multi-document context retrieval and high-precision question answering.
- Integrated OCR-based text extraction, smart document parsing, semantic chunking, and embedding generation via Hugging Face models.
- Deployed Qdrant vector database for sub-100ms vector similarity lookups, combined with Ollama LLMs for contextual response synthesis.

AI Music Studio | *Generative Audio & Music Workflows* — Python, FastAPI, LLMs, Generative AI

- Developed generative music and audio pipelines orchestrating LLMs for compositional prompts, lyric creation, and audio generation models.
- Created responsive asynchronous FastAPI endpoints to handle streaming inference, job queueing, and multi-track audio generation tasks.

VFX Artist Tracking System | *Production Operations Platform* — Python, FastAPI, PostgreSQL, REST APIs

- Engineered an end-to-end production tracking platform for media operations, tracking VFX artists, shot milestones, task statuses, and working activity.
- Designed relational PostgreSQL database schemas with optimized query indexes and RESTful endpoints for real-time artist collaboration.

EDUCATION

Bachelor of Technology in Computer Science & Engineering — Hyderabad, India

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Database Management Systems